SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Menisci and cartilage

1. Meniscal anatomy

- **Fibrocartilage**: about **70% water**; dry weight mostly **type I collagen (about 90%)**; proteoglycans, cells (**fibrochondrocytes**).
- **Collagen orientation: circumferential fibres (deep, dominant)** convert axial load into **hoop stresses**; **radial ("tie") fibres** resist longitudinal splitting; superficial random mesh.
- **Medial meniscus: C-shaped**, wider posteriorly, **firmly attached to the capsule and deep MCL**: **less mobile (about 2-5 mm excursion)**: **torn more often** (overall). **Lateral meniscus: more circular (O-shaped)**, covers more of the plateau, **more mobile (about 9-11 mm)**; attached to the **popliteus** (popliteal hiatus: no capsular attachment there) and **meniscofemoral ligaments (Humphrey, Wrisberg)**.
- **Blood supply: medial, lateral and middle genicular arteries** via a **perimeniscal capillary plexus**; only the **peripheral 10-30% (medial) and 10-25% (lateral)** is vascular in adults.
  - **Red-red zone** (peripheral, **under 3 mm from the rim**): good healing.
  - **Red-white zone** (**3-5 mm**): moderate healing.
  - **White-white zone** (**over 5 mm**, inner): avascular, poor healing.
- **Function: load transmission** (**about 50% of load in extension and about 85-90% in flexion through the menisci**; **lateral meniscus bears about 70% of lateral compartment load** vs medial about 50%), **shock absorption**, **joint stability** (**posterior horn of the medial meniscus is a secondary restraint to anterior tibial translation**, especially in ACL-deficient knees), **lubrication and nutrition**, proprioception.

- **Total meniscectomy increases contact stresses by about 2-3 times (up to 235%)**: **Fairbank changes** on radiographs after meniscectomy: **ridge (osteophyte) formation, flattening of the femoral condyle, joint space narrowing**. **Lateral meniscectomy** has worse outcomes than medial (lateral compartment convex-on-convex geometry).

2. Meniscal tears

- **Patterns:**
  - **Longitudinal (vertical): bucket-handle** (displaced longitudinal: **locked knee**; **double PCL sign** on MRI, **"fragment in the notch" sign**; commoner medially),
  - **Radial** (transects circumferential fibres: **loss of hoop stress function**; **"ghost meniscus"** sign),
  - **Horizontal (cleavage)** (degenerative; **parameniscal cysts**, especially lateral),
  - **Oblique (flap/parrot-beak)**, **complex/degenerative**,
  - **Root tears** (within about 1 cm of the root attachment): **functionally equivalent to total meniscectomy**; **medial posterior root tears in middle-aged women** with a "pop" on squatting (**meniscal extrusion over 3 mm** on MRI); **lateral posterior root tears with ACL injuries**,
  - **Ramp lesions** (posterior medial meniscocapsular junction; ACL-associated),
  - **Meniscal extrusion** (over 3 mm: root tear or degeneration).
- **Clinical:** joint-line tenderness (**most sensitive sign**), **McMurray test**, **Thessaly test (at 20 degrees)**, **Apley grind**; locking, effusion (delayed, hours).
- **MRI: sensitivity about 90-95%** (medial better than lateral); **grade 3 signal (intrasubstance signal reaching the articular surface)** = tear; grades 1-2 are intrasubstance degeneration (not tears).

3. Management of meniscal tears

- **Degenerative tears in middle-aged/older knees** (with or without OA): **physiotherapy first**; **arthroscopic partial meniscectomy (APM) is no better than sham surgery or physiotherapy** for degenerative tears (**FIDELITY**, **ESCAPE**, Katz **MeTeOR**); APM is reserved for **true mechanical symptoms (locking)** or failure of conservative care.
- **Repair indications ("save the meniscus"): peripheral (red-red/red-white), longitudinal, vertical tears 1-4 cm**, **young patient**, **acute**, **stable knee or with concurrent ACL reconstruction** (**healing rates higher with ACL reconstruction: bleeding, growth factors**), **root tears**, **ramp lesions**, **bucket-handle tears** in young patients (reduce and repair).
- **Repair techniques:**
  - **Inside-out** (**gold standard** for mid-body/posterior tears; vertical mattress sutures; **posteromedial approach (protect the saphenous nerve)**; **posterolateral approach: protect the peroneal nerve (retract biceps femoris posteriorly, work anterior to the lateral gastrocnemius)**),
  - **Outside-in** (anterior horn tears),
  - **All-inside** (posterior horn; device-based; **risk to popliteal neurovascular structures in the lateral posterior horn**),
  - **Vertical mattress sutures are biomechanically strongest** (capture circumferential fibres).
  - **Biological augmentation:** marrow venting (notch microfracture), **fibrin clot**, rasping/trephination.
- **Root repair: transtibial pull-out** repair or suture anchor; slows extrusion and OA progression vs partial meniscectomy (in knees without advanced OA).
- **Rehabilitation after repair:** protected weight bearing/limited deep flexion (**avoid squatting beyond 90 degrees for about 3-4**

months).

- **Outcomes: repair success about 70-90%** (better laterally, with ACL reconstruction, acute, peripheral); **failure rate higher medially and in isolated repairs**.
- **APM outcomes:** good for symptomatic unstable flaps in younger knees, but **long-term OA risk** proportional to the amount removed.

4. Discoid meniscus

- **Lateral (almost always)**; **about 3-5% of the population (more in Asians, up to about 15%)**; **bilateral in about 20%**.
- **Watanabe classification:**
  - **Type I: complete** (covers the entire plateau),
  - **Type II: incomplete** (covers under 80%),
  - **Type III: Wrisberg variant** (**no posterior meniscotibial (coronary) attachment**, only the **meniscofemoral ligament of Wrisberg**: **hypermobile**: **"snapping knee syndrome"** in children).
- **Imaging:** radiographs: **widened lateral joint space**, squared lateral condyle, cupping of the lateral plateau; **MRI: three or more contiguous 5-mm sagittal slices showing a continuous "bow-tie"** (meniscal body continuity), or **transverse diameter over 15 mm or over 20% of tibial width**.
- **Treatment: asymptomatic: observe**; **symptomatic: arthroscopic saucerisation (partial meniscectomy to a rim of about 6-8 mm)** ± **repair of peripheral instability (Wrisberg type)**; **avoid total meniscectomy** (early lateral OA, osteochondritis dissecans of the lateral femoral condyle).

5. Meniscal allograft transplantation (MAT) and scaffolds

- **Indication: young (under about 50), symptomatic post-meniscectomy knee** (pain in the meniscus-deficient compartment)

with minimal arthritis (Outerbridge under III), **normal alignment (or corrected by osteotomy)**, **stable knee (or concomitant ACL reconstruction)**, **BMI under about 35**.

- **Contraindications:** advanced OA, uncorrected malalignment/instability, inflammatory arthritis, obesity.
- **Size-matched fresh-frozen allograft** (radiographic sizing: **Pollard method**); fixation with **bone plugs (medial) or a bone bridge/keyhole (lateral: horns close together)**, or soft-tissue sutures.
- **Outcomes: about 70-80% survival at 10 years**; better laterally; **shrinkage, extrusion, tears** common.
- **Collagen (CMI) and polyurethane (Actifit) scaffolds** for partial segmental defects (intact rim needed).

6. Articular cartilage: structure

- **Hyaline cartilage: about 65-80% water**; **type II collagen (about 90-95% of collagen; tensile strength)**; **proteoglycans (aggrecan: GAG chains of chondroitin and keratan sulphate bound to hyaluronic acid by link protein)**: **negative charge attracts water: compressive stiffness (swelling pressure)**; **chondrocytes (about 1-5% of volume)**: produce matrix; **avascular, aneural, alymphatic**: **nutrition by diffusion from synovial fluid**: poor healing.
- **Other collagens: type IX** (cross-links type II), **type X (calcified zone, hypertrophic chondrocytes)**, **type XI** (regulates fibril size), **type VI** (pericellular).
- **Zones:**
  - **Superficial (tangential):** collagen **parallel** to the surface (**highest collagen, resists shear**), **lowest proteoglycan**, flattened cells, **lamina splendens**, **highest water content**,
  - **Middle (transitional):** oblique collagen, round cells,

  - **Deep (radial):** collagen **perpendicular** to the surface, **highest proteoglycan, lowest water**, columns of cells (resists compression),
  - **Tidemark** (separates uncalcified from calcified cartilage),
  - **Calcified cartilage** (type X collagen; anchors to subchondral bone).
- **Lubrication: boundary lubrication (lubricin: proteoglycan 4)**, **fluid-film**, **weeping/elastohydrodynamic**; **biphasic** (solid + fluid) viscoelastic material.
- **Ageing vs OA:**

Parameter	Ageing	Osteoarthritis
Water content	**Decreases**	**Increases (early)**
Collagen	Unchanged/cross-linking increases	Network disrupted
Proteoglycan content	Decreases	**Decreases**
Chondroitin sulphate	**Decreases**	**Increases (chondroitin 4-sulphate)**
Keratan sulphate	**Increases**	**Decreases**
Modulus of elasticity	**Increases (stiffer)**	**Decreases**
Degradative enzymes (**MMP-1, -13, ADAMTS-4/5**, IL-1, TNF)	Normal	**Increased**
Chondrocyte number	Decreases; size increases	Early proliferation (clusters/cloning)

7. Cartilage injury and repair

- **Partial-thickness injuries (not crossing the tidemark): do not heal** (no access to marrow cells).

- **Full-thickness (through the subchondral plate):** marrow cells form **fibrocartilage (type I collagen)**: mechanically inferior to hyaline.
- **Classification: Outerbridge** (grade I softening; **II fissuring/fragmentation under 1.5 cm (0.5 inch) diameter**; **III over 1.5 cm**; **IV exposed subchondral bone**), **ICRS** (0 normal; 1 superficial; 2 **under 50% depth**; 3 **over 50% depth (to calcified layer)**; 4 to subchondral bone).
- **Treatment algorithm for focal chondral defects** (young patients, **correct alignment, stability and meniscal deficiency first**):

Defect	Low-demand	High-demand
Small (under 2 cm^2)	**Debridement/chondroplasty**, **microfracture/marrow stimulation**	**Osteochondral autograft transfer (OATS/mosaicplasty)**, microfracture
Large (over 2-4 cm^2)	Microfracture (less durable), **ACI/MACI**	**Autologous chondrocyte implantation (ACI/MACI)**, **osteochondral allograft (OCA)** (especially with bone loss)

- **Microfracture (Steadman): debride to a stable vertical rim**, **remove the calcified cartilage layer**, awls **3-4 mm apart, 2-4 mm deep**; **fibrocartilage**; **protected weight bearing and CPM for 6-8 weeks**; **results deteriorate after about 2-5 years**, best in **young (under 40), small (under 2-4 cm^2), femoral condyle** lesions; **intralesional osteophytes (subchondral bone overgrowth)** complicate later ACI.
- **OATS (osteochondral autograft): hyaline cartilage** transfer from **less weight-bearing areas** (**lateral trochlea, intercondylar notch**); **defects under about 2 cm^2**; **donor site morbidity** (patellofemoral pain); plugs must be **flush** (**proud plugs: high contact pressures**; recessed: fibrocartilage overgrowth).

- **ACI/MACI: two-stage** (**arthroscopic biopsy of cartilage**, culture, **implantation at about 4-6 weeks** under a **periosteal patch (first generation: hypertrophy commonest complication)**, **collagen membrane (second)** or **matrix-seeded (MACI, third generation)**); **hyaline-like** repair tissue; **larger defects (2-10 cm^2)**; better than microfracture for larger defects in some trials; **slow rehabilitation (about 12 months)**.
- **Osteochondral allograft (fresh, refrigerated, used within about 28 days)**: **large defects with subchondral bone loss**, **OCD**, AVN, salvage; **viable chondrocytes (over 70% viability required)**; **no immune rejection of cartilage (chondrocytes protected in matrix)**, bone portion incorporates via **creeping substitution**.
- **Subchondral insufficiency fracture of the knee (SIFK, formerly SONK): elderly women**, **medial femoral condyle**, often after a **medial meniscus posterior root tear**; **sudden pain**; MRI: **subchondral low-signal line with bone marrow oedema**; protected weight bearing; UKA/TKA if collapse.

8. Osteochondritis dissecans (OCD)

- **Subchondral bone lesion** (repetitive microtrauma, vascular, genetic) with **risk of instability of the overlying cartilage**; **lateral aspect of the medial femoral condyle (about 70%)** (**Wilson sign**: pain at 30 degrees of flexion with internal rotation, relieved by external rotation).
- **Juvenile OCD (open physes)** has a **much better prognosis** (**about 50-75% heal with non-operative care**: activity restriction 3-6 months) than **adult OCD** (closed physes: usually need surgery).
- **MRI signs of instability (De Smet/Kijowski): high T2 signal line behind the fragment**, **cysts** beneath the lesion, **articular fracture line**, **fluid-filled osteochondral defect**.
- **Treatment:**

  - **Stable lesions in juveniles: activity modification, immobilisation**; **retrograde/antegrade drilling** if not healing after 3-6 months.
  - **Unstable but salvageable: fixation** (**bioabsorbable or metal compression screws, chondral darts**, bone pegs).
  - **Unsalvageable/loose bodies: excision + cartilage restoration (OATS, OCA, ACI)**.
- **Other sites: capitellum** (**throwers/gymnasts**: see elbow notes), **talar dome** (**posteromedial (deeper, cup-shaped, atraumatic)** and **anterolateral (shallow, wafer-shaped, traumatic)**).

Exam pearl: **Hoop stresses via circumferential fibres; radial and root tears abolish them (root tear = total meniscectomy). Red-red zone (under 3 mm) heals; repair with ACL reconstruction heals best; vertical mattress strongest. Degenerative tears: physio first (FIDELITY, ESCAPE).**

Exam pearl: **Discoid lateral meniscus: Watanabe III (Wrisberg) is hypermobile: saucerise + repair. MAT: young, meniscectomised, minimal OA, aligned, stable.**

Exam pearl: **Cartilage: type II collagen, aggrecan; superficial zone parallel collagen/highest water; deep zone highest proteoglycan. Ageing: water down, stiffer, keratan up; OA: water UP, keratan down, chondroitin up, MMP-13/ADAMTS up. Microfracture: fibrocartilage (type I); OATS: hyaline, small; ACI/MACI: larger defects; OCA: large osteochondral defects.**

Exam pearl: **OCD: lateral aspect of the medial femoral condyle; juvenile heals non-operatively more often; high T2 line behind the fragment = unstable. SIFK: elderly women, medial femoral condyle, medial root tear.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.